

SEED STENCILS FOR DIFFUSION THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

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ABSTRACT. A diffusion coefficient can vary smoothly on the scale of the grid, jump across an interface, or change smoothly over a width δ smaller than the node spacing h . Through such a sub-grid edge, standard high-order stencils that sample the coefficient at the nodes are first order until the grid resolves the edge, with a knee at $h \approx \delta$, and the interface-aware stencils built for a jump, applied at the edge's centre, sit on an error floor of order δ and leave it once the grid resolves the edge. We keep the nodes and replace the polynomial basis of the stencils that see the edge by *seeds*. The seeds are the profiles of solutions polynomial in time, continued through the edge by ordinary differential equations along the normal or along the edge's own coordinates; in one dimension they are the classical formal powers of the Sturm–Liouville operator, and as $\delta \rightarrow 0$ they reduce to the interface-aware stencils of a jump. On scattered nodes the resulting RBF-FD stencils are fourth order at every δ tested, elliptic and parabolic, with no switch between regimes. We compare them with the coefficient treatments of diffusion practice. The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. Through an edge the grid does not resolve none of them is better than second order; the conservative scheme is exact at the one-dimensional equilibrium and, on the sinusoidal medium at $\delta = 0$, ahead of the seeds below 801 nodes, where the seeds' line is pre-asymptotic; on the scattered nodes none improves on sampling by more than a factor of 1.75 in the RMS norm, and the seeds lie 2.3 to 4.7 orders below the best of them over 1250 to 40,000 nodes.

1. INTRODUCTION

Heat conduction through a composite material is governed by

$$(1) \quad u_t = \nabla \cdot (\alpha \nabla u) \equiv Lu$$

and, at equilibrium, by $Lu = 0$, with a conductivity $\alpha > 0$ that changes sharply between materials. At the two ends of the range of media the numerical picture is settled. Where α varies smoothly on the scale of the grid, a high-order stencil samples it at the nodes and keeps its order. Where α

jumps across an interface, the temperature and the flux $\alpha \partial_n u$ are continuous while the gradient is not, a standard stencil across the interface is first order, and the remedies rebuild the stencils or the elements that the interface cuts so that they satisfy the interface conditions (Section 1.1). One of them, the interface-aware RBF-FD stencils of [Mar16, MF17b] that this paper extends, keeps fourth order on scattered nodes, elliptic and parabolic, with a few rows of nodes placed across each interface. Between the two ends lies a third case: an edge that is smooth, so that no interface condition applies, but changes over a width δ smaller than the node spacing h . Every material edge is smooth on some scale, a model that resolves an edge on a fine grid presents it as sub-grid on a coarser one, and any regularisation of a jump produces one.

A standard stencil through a sub-grid edge samples α on its two sides, sees a jump, and treats the coefficient as smooth between its nodes, which is the first-order failure of the jump case. It recovers once the grid resolves the edge. In the one-dimensional ramp problem of Section 4.2, the error of the standard fourth-order operator through a 9 : 1 edge of width $\delta = 0.01$ or 0.0025 falls by 100 to 200 times between $h = 2\delta$ and $h = \delta/2$, where a first-order line falls by four. On scattered nodes in two dimensions (Section 6.2) its elliptic error stays at 0.86 to 0.94 of its error at the jump while $h \gtrsim 6\delta$, falls to 0.081 to 0.084 of it at $h \approx \delta$ for the three widths that reach it, and converges at the fifth order of the bulk stencils below. We call the turn at $h \approx \delta$ the knee.

The interface-aware stencils do not reach this regime either. Applied with the centre of the edge as the interface and the two sides' coefficients as if δ were 0, they converge to the jump problem's solution, which is not the smooth problem's: the resistance $\int dx/\alpha$ of a tanh edge between conductivities a and b falls short of the jump's by $c\delta$, $c = (a - b) \ln(a/b)/(2ab)$, so their error sits on a floor of order δ while the grid does not resolve the edge. Once it does, the rebuilt rows keep enforcing a kink that the solution lacks, and their error grows: through a 9 : 1 edge with $\delta = 0.01$ it reaches 0.11 at 1600 nodes in the one-dimensional equilibrium of Section 4.3. On scattered nodes they beat the standard stencils only while $h \gtrsim 3\delta$, and by at most a factor of 6. Using them means knowing δ and switching them off.

This paper keeps everything about the scheme except the basis of the stencils that see the edge. In place of the monomials a stencil is made exact on *seeds*, the $t = 0$ profiles of solutions of (1) that are polynomial in time. With a constant α these profiles are the polynomials, the initial values of the heat polynomials [RW59]. In one dimension, with $L = \partial_x \alpha \partial_x$ and anchored at a stencil's evaluation point x_e , they form the chain

$$(2) \quad \phi_0 = 1, \quad \alpha \phi'_1 = \alpha(x_e), \quad L\phi_k = k(k-1) \alpha(x_e) \phi_{k-2} \quad (k \geq 2),$$

with $\phi_k(x_e) = 0$ for $k \geq 1$ and $\phi'_k(x_e) = 0$ for $k \geq 2$, integrated through the edge as a first-order system of ordinary differential equations that never differentiates α (Section 3.1). The seed ϕ_1 has a constant flux, and ϕ_2 is

the profile of a solution whose rate of change u_t is constant; in general the even seeds carry $\partial_t^k u = \text{const}$ and the odd ones $\alpha \partial_x \partial_t^k u = \text{const}$, which are the interface conditions of a jump. Read across a jump, the chain is the translated polynomial basis of the interface-aware stencils (Section 3.2), so one construction covers the jump, the sub-grid edge and the resolved edge. In one dimension it is classical (Section 1.1). It is also the construction of our companion paper for the wave equation [Mar26], specialised to a single field: the two interface conditions of diffusion, on u and on its flux, are both conditions on u , so one chain serves every stencil.

In two dimensions the seeds replace the polynomial augmentation of the interface-aware RBF-FD stencils, and the Gaussian part and the saddle-point solve stay as they were. Where α varies along the normal only, as across a straight edge, seeds polynomial in the tangential coordinate turn the chain into a triangular system of ordinary differential equations in the normal coordinate, one short march per stencil (Section 5.1). Where the edge is curved or α varies along it, a profile frozen along the normal is inconsistent at $O(1)$ through an unresolved edge, and the chain is marched in the curve's own coordinates instead (Section 5.3). A resistive layer thinner than the stencil, the insulating ring of [MF17b], needs its exact width, its own series and seeds of one degree higher that carry the flux (Section 5.4). An imperfect contact costs strong-form schemes a derivative in the jump conditions, as Lombard and Piroux observed for elastic waves [LP03, LP06]; the flux seeds add the functions of one degree higher that carry the flux, and nothing else.

We measure the seeds in one and two dimensions, on elliptic and parabolic problems, against three baselines: the standard stencils with α sampled at the nodes; the interface-aware stencils of the jump applied at the edge's centre; and the coefficient treatments of diffusion practice, which change the coefficient and keep the scheme: harmonic and arithmetic means of α over a cell (over a disc on scattered nodes), the edge widened to the grid, and, in one dimension, the conservative scheme with exact face conductances. On constant pieces the one-dimensional equilibrium cannot rank these methods, since its solution has a constant flux and lies in the span of the first two seeds, on which the seeds and the exact face conductances are both exact (Section 3.5); the one-dimensional ranking is the parabolic problem's. In two dimensions the tangential variation of the solution takes it out of that span, and the equilibrium is a real test.

Contributions.

- (1) The seeds for the diffusion operator on the unchanged scattered nodes of the interface-aware RBF-FD stencils, which they reduce to as $\delta \rightarrow 0$: the one-dimensional chain (Section 3), its two-dimensional form for a straight edge in the normal coordinate, for a curved or tangentially varying edge in the curve's own coordinates, and for a resistive layer thinner than the stencil (Section 5). To our knowledge

no earlier scheme builds the basis of stencils on an unchanged scattered node set through a smooth material edge thinner than the node spacing by marching ordinary differential equations; its ingredients (the exact face conductance, the formal powers, bases from local solutions of the equation) are each classical, and in one dimension so is the construction.

- (2) In one dimension (Section 4): the knee of the standard operator, a drop of 100 to 200 times between $h = 2\delta$ and $h = \delta/2$ at a 9 : 1 edge and of 800 to 1000 times at the 10 : 1 edges of the sinusoidal medium; the jump construction on its floor, the distance of order δ between the jump's solution and the smooth one, while $h \gtrsim 2\delta$ on the two-constant medium, and growing once the grid resolves the edge; and the seeds, fourth order at every δ on one line: on the ramp problem of the two-constant medium they lie within 0.1 to 0.3%, 1 to 1.5% and 3 to 5% of the jump's line at $\delta = 0.0025$, 0.01 and 0.04.
- (3) On scattered nodes through a flat band (Section 6): the seeds are fourth order at every δ tested, elliptic and parabolic, with elliptic fits of 4.23 to 4.48 over 1250 to 40,000 nodes; the elliptic errors at the five widths, the jump included, differ by factors of 1.2 to 2.4 at each node count, and at 40,000 nodes and $\delta = 0.0025$ the seeds' error is 7.99×10^{-9} against the standard operator's 8.36×10^{-5} . That is no deterioration as $\delta \rightarrow 0$ on these node sets, not a proven δ -uniform constant. The rule is to seed every row whose stencil sees an edge within 20δ , with no threshold in δ . The seeded rows' least diagonal dominance never falls below the jump-aware rows', the seeded matrices' condition estimates and GMRES counts stay bounded as $\delta \rightarrow 0$, and their spectra have no eigenvalue in the right half-plane.
- (4) Through the curved band of case 2 the tangential chain is fourth order, within 1.4 to 3.9 times of the flat seeds' RMS error from 5000 nodes at every δ (Section 6.5). On the insulating ring thinned at a fixed resistance from $s = 10^3$ to 10^{11} the seeds are exact on the matched radial profile to rounding (1.3×10^{-15} to 2.1×10^{-15}) and converge at fourth order (fits 4.43 to 4.71) at every s ; through a smooth ring whose width and edges are both below h they converge at fourth order on the nodes away from it (Section 6.6).
- (5) The comparison with the coefficient treatments (Sections 4.5 and 6.7). The comparison is against coefficient sampling on the same nodes and against our implementation of the cell harmonic and arithmetic means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, on one contrast and one test problem per dimension. Through an edge the grid does not resolve none of them is better than second order; the conservative scheme is exact at the one-dimensional equilibrium and, on the sinusoidal medium at $\delta = 0$, ahead of the seeds below

801 nodes, where the seeds' line is pre-asymptotic; on the scattered nodes none improves on sampling by more than a factor of 1.75 in the RMS norm, and the seeds lie 2.3 to 4.7 orders below the best of them over 1250 to 40,000 nodes.

- (6) An open implementation in Python, written with the AI tools disclosed at the end of the paper, with a test for every numerical routine, in which every figure and table of this paper is drawn by a committed script from committed results files (<https://github.com/bradleypmartin/rbf-elliptic-parabolic-interfaces-2026>).

1.1. Relation to prior work. In one dimension the construction is classical in every part, and we claim nothing for it. The first seed, the constant-flux function $\alpha_e \int d\xi/\alpha$, is over one cell the exact face conductance of Tikhonov and Samarskii's homogeneous schemes [TS62, Sam01] and, for a piecewise-constant coefficient, the harmonic-mean interface conductivity of finite-volume heat conduction [Pat80]; the conservative scheme built on it, exact at the source-free equilibrium, is second order [PXL26]. The whole chain is the representation of Sturm–Liouville solutions by recursive integrals, as power series in the spectral parameter [Kra08, KP10], taken from the anchor; for a constant coefficient its time-polynomial solutions are the heat polynomials [RW59, Wid75]. Three-point schemes of any order for $(ku)'$ with a piecewise-smooth k , whose coefficients come from Cauchy problems integrated across each cell, are the exact and truncated difference schemes [SM90, GHMK11]; three-point schemes of any order exist across a jump too [Gar89], one of them built by carrying Taylor data across the step [Suj10]. In two dimensions, local solutions of the equation that satisfy the interface conditions are the basis of the Trefftz difference schemes of FLAME, on irregular stencils too [Tsu06, Tsu10], and of the tailored finite point methods [Hua09, HZ12]; Galerkin methods build their bases from local solves [BCO94, HW97, CGH10, OZ07, MP14] and reach high order through coefficients the mesh does not resolve [Mai21, HP23], and the quasi-Trefftz spaces, one of them for the heat equation, build polynomial local solutions by Taylor expansion [IGMPS25, GPS26], which cannot reach through an edge thinner than the stencil [Mar26]. For a jump, the interface-aware RBF-FD stencils this work generalises [Mar16, MF17b] stand beside the immersed interface method and unfitted finite elements [LL94, LI06, HH02, BCH⁺15], now of fourth and sixth order [FZ20, FHM24], all of which take the jump as given. A resistive layer thinner than the grid is an imperfect interface in the limit [SP74, PP13, Has01], which Section 5.4 takes up. The standing alternative changes the coefficient rather than the stencil: harmonic or arithmetic means over a cell [Pat80, EIL01, GA92], averaging for layering finer than the grid [MDH⁺99, DDH03], or a regularised coefficient [TE03, TE06]. Those methods take the cell or the width from the grid; here the width is a property of the medium, below the node spacing, and the aim is the scheme's full order through it. We found no prior instance of stencils

on an unchanged scattered node set whose basis is continued through a sub-grid smooth edge by ordinary differential equations, nor of such stencils reducing to a jump construction as $\delta \rightarrow 0$; the search is logged in the repository’s `LITERATURE.md`. We ran the cell means, a widened edge and, in one dimension, the conservative scheme with exact face conductances, as our implementation of each on one problem per dimension: through an edge the grid does not resolve none is better than second order, and on the scattered nodes none improves on sampling the coefficient by more than a factor of 1.75 in the root-mean-square error. We claim nothing beyond that measurement.

Outline. Section 2 fixes the problems, the discretisations and the interface-aware stencils of a jump that the seeds extend. Section 3 constructs the seeds in one dimension and proves what can be proved about them. Section 4 measures them against the standard operator, the jump construction and the coefficient treatments. Section 5 builds the seeds on scattered nodes for straight, curved and doubly sub-grid features. Section 6 measures them there, elliptic and parabolic, with the matrices they make. Section 7 records the limitations and what we would do next.

2. SETTING

Notation. The symbols the sections share, fixed here:

n, h	node count and spacing: $h = 2/(n - 1)$ in one dimension and the row spacing $1/\text{round}(0.95\sqrt{n})$ in two
$\alpha; \alpha^-, \alpha^+; L$	conductivity; the pieces on the two sides of an edge; $L = \nabla \cdot (\alpha \nabla \cdot)$
$\delta; \sigma$	width of a tanh edge, $\delta = 0$ the jump; the blend weight $\sigma(z) = \frac{1}{2}(1 + \tanh z)$
s	the thinness of the ring of [MF17b], $1/s$ its width
$x_e; (x', y')$	a stencil’s evaluation point (anchor); its local frame, y' normal to the edge
ϕ_k	the one-dimensional seeds (2)
$D_x, D_y; A$	first-derivative matrices; $\text{diag } \alpha$ at the nodes
naive, direct	$D_x A D_x (+D_y A D_y)$; $\alpha \nabla^2 u + \nabla \alpha \cdot \nabla u$ on one stencil
jump-aware	the interface-aware stencils of Section 2.4
the $\delta = 0$ construction	the same, applied at a smooth edge’s centre
seeds	the operator whose rows that see an edge are rebuilt on seeds

2.1. The problem and its interface conditions. On a domain Ω in one or two dimensions we solve (1) with Dirichlet data on $\partial\Omega$, and its equilibrium $Lu = 0$, the elliptic problem. The conductivity $\alpha > 0$ is one smooth function on each side of every interface Γ . Across a jump the temperature and the heat flux are continuous at all times, hence so are all their time derivatives,

and $\partial_t^k u = L^k u$ turns these into conditions on spatial derivatives,

$$(3) \quad [L^k u] = 0, \quad [\alpha \partial_n L^k u] = 0 \quad \text{on } \Gamma, \quad k = 0, 1, 2, \dots,$$

where $[\cdot]$ is the difference of the two sides' limits, n the normal to Γ , and each side's L is taken with that side's α [Mar16, MF17b].

A smooth edge. With an edge width $\delta > 0$ the jump at an interface is replaced by a tanh transition between the two pieces,

$$(4) \quad \alpha = (1 - \sigma(d/\delta)) \alpha^- + \sigma(d/\delta) \alpha^+, \quad \sigma(z) = \frac{1}{2}(1 + \tanh z),$$

where d is the signed distance to the interface, $x - x_c$ in one dimension and the distance along the normal through the foot point on the curve in two, positive on the side of α^+ . The pieces keep their own variation, and $\delta = 0$ is the jump. A band between two interfaces is blended twice, from the outside piece into the inside one across the lower interface and back across the upper, so that a band wider than its edges has two independent transitions; Section 5.4 defines the smooth form of a layer thinner than its edges. Beyond 19δ from its centre an edge is its pieces to within two units in the last place.

Three regimes. For $\delta \gg h$ a stencil sees a smooth coefficient. For $\delta = 0$ it sees a jump, and (3) applies. For $0 < \delta \lesssim h$ it sees what looks to it like a jump between two of its nodes, but no interface condition applies, since α is smooth, and the smooth problem and the jump problem have different solutions. They differ at first order in δ : in one dimension the resistance $\int dx/\alpha$ of a tanh edge between constant pieces a and b falls short of the jump's by

$$(5) \quad c\delta, \quad c = \frac{(a-b)\ln(a/b)}{2ab},$$

which is 8.79δ for the 9 : 1 edge of Section 2.2 and is positive for either ordering of a and b .

Norms and orders. In one dimension an error is the relative ℓ^2 error at the nodes, $\|e\|_2/\|u\|_2$, and a rate is $\log_2$ of the ratio of consecutive errors as the node count doubles. In two dimensions an error is the root-mean-square (RMS) error over all nodes, the Dirichlet nodes included, and an order is the least-squares slope of $\log e$ against $\log h$ over the node counts named. A maximum-norm error is named as such wherever one is quoted, and every caption says which norm it shows.

2.2. One dimension. The one-dimensional problem is (1) with $L = \partial_x \alpha \partial_x$ on $[-1, 1]$ with Dirichlet ends. A grid has n equispaced nodes with both ends on nodes, so $h = 2/(n-1)$. Two media are used, each at $\delta = 0, 0.04, 0.01$ and 0.0025 :

- the *two-constant medium*, $\alpha = 1/9$ for $x < 0$ and 1 for $x > 0$, the single interface of the time-dependent test in the code behind [Mar16], which the dissertation does not report; the edge lies midway between two nodes when n is even;

- the *sinusoidal medium* of [Mar16, eq. (75)], $\alpha = 0.1 + 0.4 \sin 2\pi x$ on the layer $[0, \frac{1}{2}]$ and 1 elsewhere, whose two edges both have the contrast 10 : 1, since the sinusoid vanishes at both; both edges fall on nodes when $n - 1$ is a multiple of 4.

Each medium carries two problems: the equilibrium with $u(-1) = 1$ and $u(1) = 0$, and the *ramp problem*, which starts from $u = 0$, holds $u(1, t) = 0$, raises $u(-1, t)$ from 0 to 1 over $0 \leq t \leq 1$ by the C^∞ step $e^{-1/t}/(e^{-1/t} + e^{-1/(1-t)})$, and is read at $t = 2$. (The original code ramps with a logistic function, whose value and derivatives at $t = 0$ are small but not zero; the mismatch with the zero start at the corner would cap a fourth-order study.) Section 4.1 gives the reference solutions.

The *naive* operator is $D_x A D_x$ [Mar16, eq. (76)], with D_x the fourth-order first-derivative matrix of Fornberg’s weights [For88], one-sided at the ends, and $A = \text{diag } \alpha(x_j)$, which gives nine-point rows in the interior. It is the form in which a standard stencil sees a jump at all: a direct discretisation of $\alpha u_{xx} + \alpha' u_x$ with the coefficients sampled at the nodes is blind to the jumps, and its error on the sinusoidal medium stays at 0.22 as the grid is refined. Time stepping is the fourth-order backward differentiation formula (BD4) with $\Delta t = h$, one sparse LU factorisation per operator and the Dirichlet values imposed at every step; its three starting values come from classical Runge–Kutta sub-steps at half the stability limit.

2.3. Two dimensions. The two-dimensional problems are the three test cases of [MF17b], (1) on the unit strip $0 \leq y \leq 1$, periodic in x with period 1, with Dirichlet data on $y = 0$ and $y = 1$. The conductivity is one smooth piece in a band between two interfaces and another outside it.

- *Case 1*, a flat band $0.6 \leq y \leq 0.8$ with $\alpha = 0.2$ in it and 1 outside. With $u = \sin 2\pi x$ on $y = 1$ and $u = 0$ on $y = 0$ the equilibrium is $\sin 2\pi x v(y)$ with v piecewise exponential, and the parabolic problem is marched to $t = 0.1$ against the exact solution $e^t \sin 2\pi x w(y)$ of the same form [MF17b].
- *Case 2*, both interfaces bent to $y = 0.6 + 0.02 \sin 2\pi x$ and $y = 0.8 + 0.02 \sin 2\pi x$,¹ with $\alpha = 0.2 + 0.1 \sin 2\pi x \sin 2\pi y$ in the band and the boundary data of case 1.
- *Case 3*, a ring $0.35 - 1/s \leq r \leq 0.35$ about $(\frac{1}{2}, \frac{1}{2})$ with $\alpha = 1/(1.5s) + \sin 2\pi x \sin 2\pi y/(3s)$ in it and 1 outside, around a cooling circle $r = 0.05$ whose inside is removed, with $u = \sin 6\pi x$ on $y = 0$ and $y = 1$ and $u = 0$ on the circle. At $s = 10^3$ this is the insulating ring of [MF17b], 0.001 wide at a contrast of about 1500 : 1; the extremizing sweep of that paper runs s up to 10^{11} , thinning the ring at a fixed resistance, its width over its conductivity, of $1.5/(1 + \frac{1}{2} \sin 2\pi x \sin 2\pi y)$.

¹The published texts do not state the curves; these are the ones of the original code.

With $\delta > 0$ each interface of cases 1 and 2 becomes an edge (4) in the signed normal distance to its curve; the ring's smooth form is Section 5.4's. On case 1 the medium is then the one-dimensional one in y , and the reference solutions stay separable (Section 6.1).

Node sets. A set of n nodes is built as in [MF17b]. Rows hold $m = \text{round}(0.95\sqrt{n})$ nodes per unit length, so their spacing is $h = 1/m$. Six fixed rows straddle each interface, three on each side at normal distances $(\frac{1}{2} + j\frac{\sqrt{3}}{2})h$, $j = 0, 1, 2$, the middle row on each side staggered by half a spacing along the curve, so that the rows cross the interface in a hexagonal pattern; a fixed row lies on each Dirichlet curve; and the remaining nodes are placed by 100 steps of a simulated repulsion from a random start, kept clear of the straddling rows. The ring is thinner than h at every node count used, so one set of six rows straddles its midline, and both of its interfaces pass between the innermost pair. Unless a spread over node sets is quoted, each node count has one node set, from a fixed random seed.

RBF-FD. RBF-FD [FF15a, FF15b] builds each row of a differentiation matrix from a stencil. For a stencil of nodes $x_1, \dots, x_{N_s}$ about an evaluation node x_c , the weights w of a linear operator $\mathcal{L}$ solve

$$(6) \quad \begin{bmatrix} \Phi & P \\ P^\top & 0 \end{bmatrix} \begin{bmatrix} w \\ w_* \end{bmatrix} = \begin{bmatrix} \mathcal{L} \varphi(\|x - x_i\|)|_{x_c} \\ \mathcal{L} p_j(x)|_{x_c} \end{bmatrix},$$

with $\Phi_{il} = \varphi(\|x_i - x_l\|)$ for the Gaussian $\varphi(r) = e^{-(\varepsilon r)^2}$, $P_{ij} = p_j(x_i)$ over the bivariate monomials p_j up to a degree p in coordinates scaled by the stencil's radius, and w_* discarded. The constraint rows make the weights exact on the polynomials, which take over from the Gaussians under refinement [FFBB16, BFFB17]; the shape parameter is $\varepsilon = 0.4/d$, with d the distance from x_c to its nearest neighbour. Stencils have 42 nodes and degree 5 in the bulk, and 30 nodes and degree 4 within $3/\sqrt{n}$ of a Dirichlet curve and wherever a 42-node stencil would reach across an interface. Of the latter, the interface-aware construction rebuilds the rows whose 30 nodes cross an interface (Section 2.4); the seeds rebuild the rows whose stencils see an edge (Section 5.2).

Operators and solvers. The naive operator is $D_x A D_x + D_y A D_y$, the two-dimensional form of [Mar16, eq. (76)], with every node's rows of D_x and D_y built on its own stencil, so that it reaches the neighbours of the neighbours. The direct operator applies $\alpha \nabla^2 u + \nabla \alpha \cdot \nabla u$ at each node through one stencil; it is blind to a jump and serves as the control where the grid resolves the edge. Dirichlet rows are the identity. The elliptic systems are solved directly with SuperLU; the iterative study applies GMRES and BiCGSTAB to the interior system, with and without an incomplete LU factorisation and the diagonal-dominance preconditioner of [Mar16, App. B]. The parabolic problem of case 1 is marched by BD4 with $\Delta t = h$ from the exact solution's history, 3 to 19 steps to $t = 0.1$.

2.4. Interface-aware stencils for a jump. The seeds extend the interface-aware stencils of [Mar16, MF17b], whose wave-equation form is in [MFSC15, MF17a]. We summarise them in the form the later sections use; the derivations are in those references.

One dimension. Expand u on each side of an interface x_c in a polynomial of degree four about x_c , with the Taylor coefficients of that side's α entering through multiplication matrices on the coefficient vectors, and impose (3) on $L^k u$ for $k = 0, 1, 2$ and on the flux for $k = 0, 1$. These five conditions give a square *continuity matrix* per side, $C^- a^- = C^+ a^+$ on the coefficient vectors $a^\pm$. Keeping the monomials on the evaluation node's side and carrying them across with $(C^+)^{-1} C^-$ gives the *translated basis*: piecewise polynomials, one per monomial, that satisfy the five conditions. The weights of a five-node stencil that straddles x_c reproduce L at the evaluation node on this basis, a 5×5 solve per stencil, and a stencil that straddles both edges of a thin layer translates twice.

Two dimensions. A stencil whose 30 nodes lie in more than one region is given a local frame at the point of each interface nearest its centre, x' along the tangent and y' along the normal, in units of the stencil's radius, and the interface is expanded there as $y' = f(x')$. The continuity conditions equate, along the interface, the first $5 - 2k$ coefficients in x' of $L^k u$ and the first $4 - 2k$ of the normal flux $\alpha(\partial_{y'} - f' \partial_{x'}) L^k u$, fifteen conditions for degree four, one per bivariate monomial; the curvature enters through f , in the curve along which the coefficients are read and in the normal derivative. The translated basis replaces the monomials p_j in (6). The Gaussians are *warped*: in the frame of the evaluation node's side, the normal coordinate of every node is replaced by a piecewise-linear one whose slope changes by the factor α^-/α^+ across each interface, so that a Gaussian's value and its $\alpha \partial_n$ are both continuous there and the RBF part of the stencil satisfies the flux condition to first order.

This construction, with warped Gaussians, is what we call *jump-aware*. Applied with the centre of a smooth edge as the interface and the pieces' Taylor data as if δ were 0, it is the $\delta = 0$ *construction* of Sections 4.3 and 6.2. It is also the $\delta \rightarrow 0$ limit of the seeds, which reproduce it at $\delta = 0$ (Sections 3.2 and 5.2).

2.5. Scope of the implementation. The experiments run on a reimplementation of the 2016–2017 methods in Python (NumPy and SciPy, with a test for every numerical routine), written for this study. Every published figure of the one- and two-dimensional heat problems of [Mar16, MF17b] but the cornered interface's has a counterpart in it, and every published convergence line is reproduced at its published rate or better. Where the numbers part company the repository's port notes say so. Two differences bear on this paper: the iterative solvers do not break down on the ring here (BiCGSTAB converges at every node count), and the extremizing sweep does not lose convergence at $s = 10^{11}$ (every s is fourth order down to its

reference's own error). The 2016 two-dimensional code is not preserved, so neither difference could be traced to it.

The implementation leaves out the cornered interface of [Mar16, Sec. 5.4.5], which is approximated there by circular segments at second order. Its two-dimensional media are a band between two curves, periodic in x , so no edge meets the domain's boundary. The bulk stencils are degree five and the stencils that see an edge degree four, seeded or interface-aware. Each case has one contrast: 9 : 1 and 10 : 1 in one dimension, 5 : 1 on cases 1 and 2, and on the ring about 1500 : 1 at $s = 10^3$, growing with s at a fixed resistance. Every smooth edge is a tanh. Section 7 returns to these restrictions.

3. SEEDS FOR THE DIFFUSION OPERATOR IN ONE DIMENSION

3.1. Seeds as the profiles of solutions polynomial in time. [Stub, #46: the chain of seeds anchored at the evaluation node, continued through the edge in first-order form so that α is never differentiated; the seed-basis figure.]

3.2. Nested kernels and the jump as the limit. [Stub, #46: the nested kernels of the operator; the translated basis of Section 2.4 as the $\delta \rightarrow 0$ limit.]

3.3. Well-posedness. [Stub, #46: the seeds form an extended complete Chebyshev system, so every stencil solve is nonsingular.]

3.4. Truncation error. [Stub, #46: third order locally on the seeded rows, fourth globally; the two readings of the bound.]

3.5. The elliptic problem in one dimension. [Stub, #46: the exactness remark and its finite-volume twin, citing the literature #43 finds.]

4. ONE-DIMENSIONAL RESULTS

4.1. Media and references. [Stub, #47: the smooth-edged media and the references at every δ .]

4.2. The parabolic knee. [Stub, #47: the naive knee at $h \approx \delta$; the snapshot of the four regimes on one grid.]

4.3. The jump construction on a smooth edge. [Stub, #47: the $\delta = 0$ construction's error on a smooth edge.]

4.4. The seeds. [Stub, #47: the seeds' rates at every δ .]

4.5. Coefficient treatments. [Stub, #47: the treatments' table and the ranking; the elliptic sanity figure as a remark.]

5. SEEDS IN TWO DIMENSIONS

5.1. A straight feature. *[Stub, #48: the local frame and the anchor; seeds polynomial in the tangential coordinate with coefficient functions marched along the normal.]*

5.2. Stencil assembly. *[Stub, #48: the seeds in the RBF-FD system, the warp they bring with them, which rows are seeded; the seed-section figure.]*

5.3. Curved and tangentially varying features. *[Stub, #48: why a frozen normal profile is inconsistent where α varies along the edge; the tangential chain in the curve's own coordinates.]*

5.4. A thin resistive layer. *[Stub, #48: the double edge of a ring: its exact width, series composition, the flux seeds, the warp on rows anchored inside the layer.]*

6. TWO-DIMENSIONAL RESULTS

6.1. Test problems and references. *[Stub, #49: the flat band, the sine pair, the ring; the separable, product-grid and reference-free references (stiff 4.1, 4.6, 4.8; statement 1).]*

6.2. Naive baseline. *[Stub, #49: the knee on scattered nodes and the jump construction on its floor (stiff 4.2; statements 2–3).]*

6.3. Solvability and spectra. *[Stub, #49: the seed rows in the global matrix (stiff 4.4; statement 6).]*

6.4. The flat sweep. *[Stub, #49: the seeds against the baselines at every (n, δ) , the warp, the elliptic ranking (stiff 4.5; statements 4, 5, 7).]*

6.5. The curved feature. *[Stub, #49: the frozen profile and the tangential chain on the sine pair (stiff 4.6–4.7; statement 8).]*

6.6. The ring and the extremizing sweep. *[Stub, #49: the twins of EABE Fig. 19–20 and the smooth ring (stiff 4.8, 4.10; statement 9).]*

6.7. Coefficient treatments. *[Stub, #49: the treatments on scattered nodes (stiff 4.9; statement 10).]*

6.8. A snapshot. *[Stub, #49: the four regimes on one node set (stiff 5.1; statement 11).]*

7. DISCUSSION

7.1. Limitations. *[Stub, #50: corners, extreme contrasts, the cost in three dimensions, what the experiments leave open.]*

7.2. Future work. *[Stub, #50: in the order to do it.]*

7.3. Conclusions. *[Stub, #50: LITERATURE.md section 6b, verbatim.]*

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[Stub, #52: acknowledgments; where the code and data live.]

TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

[Stub, #52.]

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